Olist E-Commerce

Customer, Category & Delivery Performance Analysis

An end-to-end analysis of the Olist Brazilian e-commerce dataset, covering customer segmentation, category-level sales performance, and the relationship between delivery delays and customer satisfaction.

PostgreSQL · Python (pandas, seaborn, scipy) · SQL

Introduction

Olist is a Brazilian e-commerce platform that connects small and medium-sized merchants to major online marketplaces. This analysis uses Olist's public dataset of orders, products, payments, and customer reviews between 2016 and 2018 to answer three practical business questions:

- Who are the platform's customers, and where is revenue actually concentrated?
- Which product categories drive the most sales, and how does that relate to customer satisfaction?
- Does delivery performance affect customer satisfaction, and if so, where should operational attention focus?

The technical implementation — schema design, SQL querying, and the Python cleaning/analysis pipeline — is documented separately in the project's README. This report focuses on findings and their business interpretation.

Executive Summary

- Only about **3% of customers** made more than one purchase. Repeat-purchase-based loyalty is not a meaningful lever in this dataset — retention strategy would need to be rebuilt around acquisition and one-time purchase value instead.
- Revenue is heavily concentrated in a small number of large, one-off transactions rather than loyal repeat buyers. The largest revenue segment by volume (**“Lost”**, customers inactive for a long time) still accounts for **38% of total revenue** simply due to its size, while the **“Potential”** segment (15% of customers, 28% of revenue) shows a much stronger revenue-per-customer ratio — making it the highest-leverage segment for re-engagement.
- The top 5 product categories generate a disproportionate share of revenue, but the highest revenue category is not the highest rated — category-level pricing and fulfillment quality vary independently of sales volume.
- Delivery delay has a statistically significant, negative relationship with review scores (Spearman rho ~ -0.15, $p < 0.001$). Practically, this shows up sharply at the extremes: when an order arrives late, nearly **47% of reviews are 1-star**, compared to just 22% being 5-star.
- Late delivery risk is not evenly distributed — the **Northeast region has roughly double the late-delivery rate of the Southeast** (13.9% vs. 7.1%), and a small number of sellers carry disproportionately high late-delivery rates.

1 Customer Segmentation (RFM)

Question

Who buys from Olist, how often, and how much revenue does each type of customer represent? Can customers be grouped into actionable segments?

Method

Recency, Frequency, and Monetary (RFM) values were calculated per unique customer from delivered orders. Because repeat purchases are rare, frequency was treated as a binary signal (single vs. repeat buyer) rather than a fine-grained count. The top 1% of customers by spend were separated out as high-value outliers before segmenting the remaining customer base into eight behavioral segments (e.g. Champion, Loyal, Potential, Recent, At Risk, Lost).

Findings

Repeat purchase behavior is rare: only **2.99%** of customers bought more than once. This single fact shapes the rest of the segmentation — most customers can only be evaluated on recency and spend, not loyalty.

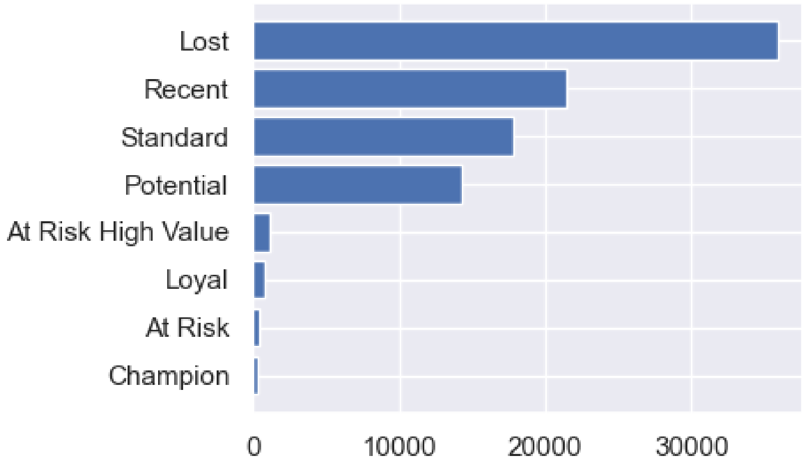

Figure 1.1 — Customer count by RFM segment

The customer base skews heavily toward inactive or low-engagement segments: **“Lost”** customers alone make up **38.9%** of the base, followed by **“Recent”** (23.3%) and **“Standard”** (19.4%). High-value, engaged segments (Champion, Loyal, At Risk High Value) together represent only about **2.5%** of customers.

Segment size does not track with segment value. “Champion” customers spend the most per person (avg. R$364) and purchased most recently (avg. 46 days ago), but there are only 325 of them — under 1% of revenue. Revenue is instead concentrated where volume is largest: “Lost” customers generate **38.1% of total revenue** and “Potential” customers generate **28.1%**, together accounting for two-thirds of all revenue despite low individual engagement. The table below breaks this down segment by segment.

Segment	Customers %	Revenue %	Avg. Spend (R$)	Avg. Recency (days)
Lost	38.94%	38.12%	128.22	392.67
Potential	15.48%	28.07%	237.53	91.98
Standard	19.36%	18.45%	124.81	219.77
Recent	23.25%	9.94%	56.01	89.95
At Risk High Value	1.27%	2.76%	283.74	313.89
Loyal	0.91%	1.44%	206.34	105.67
Champion	0.35%	0.98%	364.10	46.13
At Risk	0.42%	0.24%	75.19	333.94

Table 1.1 — RFM segment overview, sorted by revenue share

Figure 1.2 — Cumulative customer share vs. cumulative revenue share (Pareto curve)

The Pareto curve sits close to the diagonal reference line rather than bowing sharply above it. This confirms visually what the segment breakdown suggests: Olist's revenue base is **broad and comparatively flat** rather than driven by a small core of loyal high-spenders — a structurally different (and more fragile) shape than the classic “80/20” retail pattern.

What this means

Because revenue is not anchored by a loyal core, Olist's revenue base is more exposed to swings in new-customer acquisition than a typical repeat-purchase retailer would be. The “Potential” segment — recent, reasonably high-spending customers who haven't yet returned — stands out as the highest-leverage group for retention campaigns: it already carries real revenue weight (28%) and, unlike “Lost” customers, is still recently active enough to realistically re-engage.

Recommended actions:

• Launch a targeted re-engagement campaign (personalized discount, reminder email) aimed specifically at the “Potential” segment — it combines meaningful revenue share with recent activity, making it realistically recoverable.

• Deprioritize broad win-back spend on the “Lost” segment relative to its revenue share; a low-cost automated channel is more appropriate than resource-intensive campaigns.

• Track “Potential to Loyal” conversion rate over time as a KPI for retention initiatives.

2 Category Performance

Question

Which product categories drive Olist's sales, and does higher revenue come at the cost of lower customer satisfaction?

Method

Orders were grouped by product category to calculate total revenue, units sold, average product price, and average review score per category. A market-basket / association-rule analysis was initially planned to look at cross-category purchase patterns, but was scoped out after the data showed it wouldn't be informative (see below) — a deliberate, data-driven decision rather than an incomplete analysis.

Findings

Over **99% of orders contain products from a single category** (95,433 of 96,211 orders); only 760 orders span two categories and 18 span three. With cross-category co-purchases this rare, association-rule mining would not surface meaningful patterns, so it was dropped from scope rather than forced.

Revenue is led by **health_beauty** (R$1.23M), **watches_gifts** (R$1.16M), **bed_bath_table** (R$1.02M), **sports_leisure** (R$953K) and **computers_accessories** (R$888K).

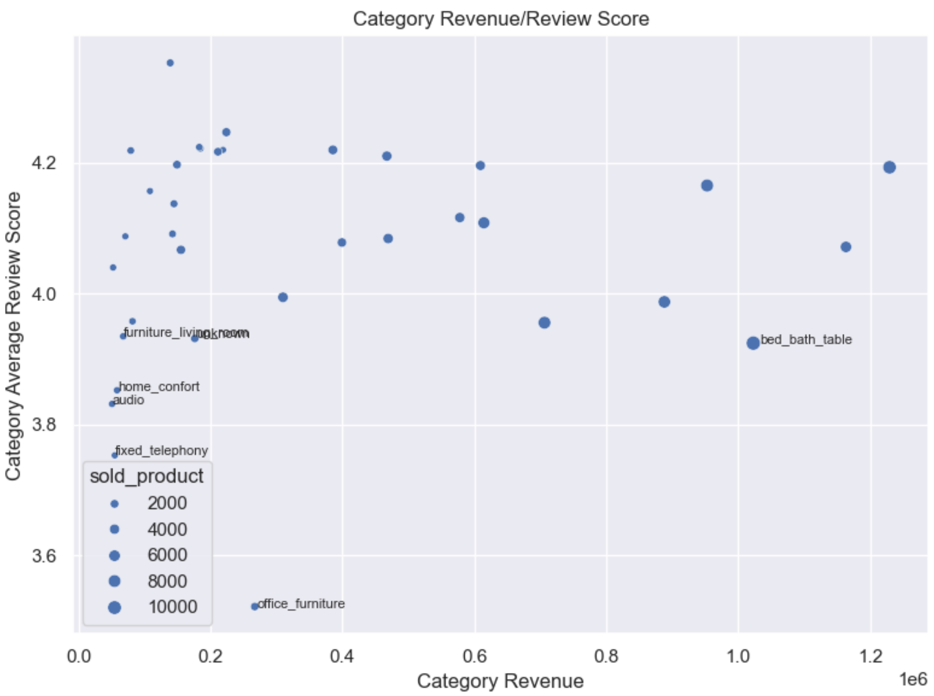

Figure 2.1 — Revenue vs. average review score for top categories by revenue (bubble size = units sold)

Looking at **watches_gifts** specifically — the second-largest category by revenue — monthly revenue shows a sustained upward trend across the dataset's timeframe rather than a single event-driven spike. The category's highest month is **May 2018 (~R$119K)**, roughly 25% above its next-highest month (November 2017, ~R$95K). The overall growth pattern, rather than any one month, is the more reliable signal here.

Figure 2.2 — Watches & Gifts: monthly revenue trend

Revenue and satisfaction are largely independent of each other. Among the top revenue categories, **luggage_accessories** has the highest average review score (4.35) despite generating only ~1% of total revenue, while some higher-revenue categories sit closer to the 3.9–4.0 range. At the other end, **security_and_services** (avg. 2.50), **diapers_and_hygiene** (avg. 3.33) and **office_furniture** (avg. 3.52) stand out as categories with a satisfaction problem independent of their sales volume.

What this means

High-revenue categories aren't automatically the ones customers are happiest with, and low-rated categories aren't necessarily the low-volume ones businesses can afford to ignore. Category-level quality issues (fulfillment, product-fit expectations, packaging fragility for categories like security equipment or furniture) look like a separate lever from simply growing sales volume, and would need to be addressed category-by-category rather than platform-wide.

Recommended actions:
• Prioritize a quality/fulfillment review for **security_and_services**, **diapers_and_hygiene**, and **office_furniture** — their satisfaction gap is not explained by low sales volume alone.
• Investigate what drove **watches_gifts**'s May 2018 peak (campaign, pricing, seasonal gifting cycle) — its sustained upward trend suggests the category has room to grow further if the driver can be identified and repeated.
• Track category-level review score as a separate KPI from revenue in category management dashboards, rather than assuming the two move together.

3 Delivery Delay & Customer Satisfaction

Question

Does the gap between estimated and actual delivery dates affect how customers rate their orders? If so, where geographically should operational improvements focus?

Method

Delivery delay was calculated as the difference between actual and estimated delivery dates for each order, then compared against the review score left for that order. Because review scores are ordinal (1–5) rather than continuous, the relationship was tested using **Spearman rank correlation** rather than Pearson's. Regional and state-level late-delivery rates were calculated separately, applying a minimum order-count threshold per state (≥300) and per seller (≥10 reviews) to avoid small-sample noise.

Findings

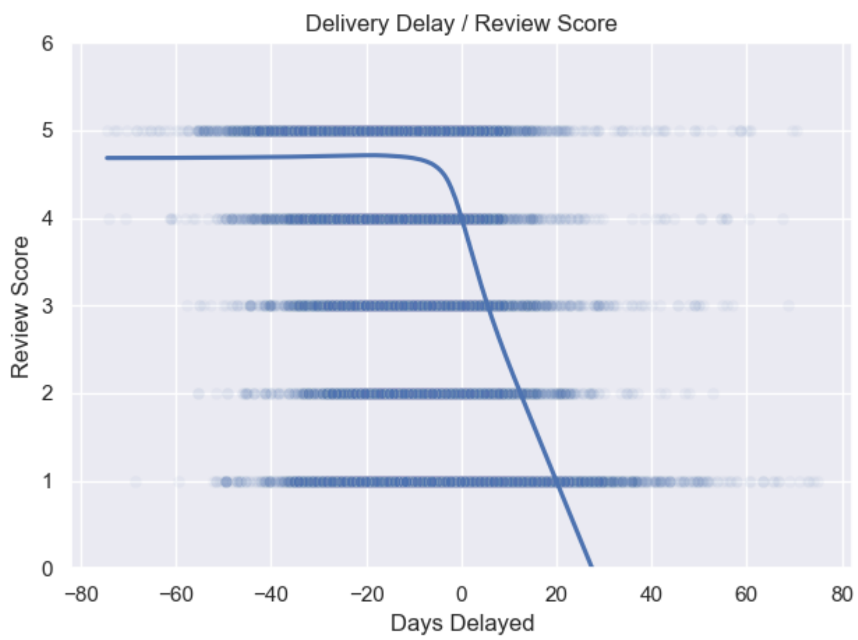

Figure 3.1 — Delivery delay (days) vs. review score, with LOWESS trend line

There is a statistically significant negative correlation between delivery delay and review score (**Spearman rho ~ -0.152, p < 0.001**). The relationship is modest in strength but highly consistent given the sample size, and becomes much more visible when looking specifically at late orders: among orders that arrived after the estimated date, **47.0% received a 1-star review**, compared with just **22.1% receiving 5 stars** — a near-inversion of the typically dominant 5-star pattern seen in most review datasets.

Late-delivery rates vary substantially by region: the **Northeast (13.9%)** and **North (9.3%)** see meaningfully higher late-delivery rates than the **Southeast (7.1%)** and **South (6.7%)**. At the state level, **Alagoas (AL, 23.6%)**, **Maranhão (MA, 19.9%)** and **Sergipe (SE, 15.8%)** stand out as the highest-risk states. At the seller level, a small number of sellers show late-delivery rates above

40–60%, well beyond the platform average.

Region	Late-Delivery Rate
Northeast	13.9%
North	9.3%
Midwest	7.8%
Southeast	7.1%
South	6.7%

Table 3.1 — Late-delivery rate by region, sorted descending

What this means

Delivery reliability behaves less like a general satisfaction driver and more like a **dissatisfaction trigger** — its effect is concentrated at the negative end (turning 5-star experiences into 1-star ones) rather than uniformly shifting the whole rating scale. This suggests operational fixes should prioritize **reducing the incidence of lateness** (especially in the Northeast, and for the specific sellers with the worst records) over trying to improve satisfaction broadly — the highest-leverage fix is preventing the worst outcomes, not polishing the average one.

Recommended actions:
• Prioritize logistics/operational investment (carrier selection, warehouse placement) in the **Northeast region** given its outsized late-delivery rate.
• Set up seller-level monitoring with an alert threshold (e.g. >20% late-delivery rate) to flag the small number of consistently underperforming sellers identified.
• Frame delivery KPIs around **reducing late-delivery incidence** rather than improving average delivery time, since the effect on satisfaction is concentrated at the negative extreme.

Limitations

- The dataset covers 2016–2018 only; findings reflect that specific window and may not hold for Olist's current operations.
- The order_reviews table has no primary key in the source data, which required manual deduplication logic (kept the latest response per order) rather than a guaranteed-clean join.
- With only ~3% of customers making a repeat purchase, frequency-based and cohort/retention analysis has limited statistical depth compared to datasets with stronger repeat-purchase behavior.
- Market basket / association-rule analysis was scoped out, as described in Section 2, due to the near-absence of multi-category orders — this is a deliberate scope decision, not a gap in the pipeline.
- Correlation is not causation: the delivery-delay/review relationship is a statistical association, not a controlled experiment, and other unmeasured factors (product issues, customer service response) could also contribute to review scores.

Conclusion

Across all three analyses, a consistent theme emerges: Olist's business is built on breadth rather than depth. Revenue comes from a wide base of largely one-time purchasers rather than a loyal core, sales are spread across many product categories rather than concentrated ones, and satisfaction is shaped more by avoiding operational failures (late delivery) than by maximizing an already-decent average experience. The highest-leverage opportunities identified here — re-engaging the “Potential” customer segment, addressing category-specific satisfaction gaps, and reducing late deliveries in the Northeast and among the worst-performing sellers — are all narrower and more targeted than a platform-wide initiative would be, and reflect where the data suggests the most impact per unit of effort.